

Volume Explorer — Rectangular Prisms

Name: _____ Date: _____ Class: _____

Build, observe, calculate, and explain.

Geometry World uses the familiar building and exploration of block-based 3D sandbox games to make geometric measurement visible. Count cubes in a layer, connect layers to multiplication, and compare strategies. Explaining and checking those calculations provides practice toward accurate, flexible calculation fluency.

Measure and calculate the volume of 3D shapes by exploring, building, and counting blocks.

Lesson goals

1. Find the volume of the blue rectangular prism ($5 \times 3 \times 4$)
2. Fill the empty pool with blocks and count them
3. Calculate the volume of the L-block

Your lesson route

1. Professor Block · investigation
2. Quiz Master · 4 question steps
3. Builder Bot · 2 question steps
4. L-Block Sage · 3 question steps

Print in portrait at 100% or Save as PDF. This companion follows the current lesson's tasks and question choices. Answer by hand; a drawing, words, or another clear representation can show your thinking. Use the world to gather observations before completing the paper independently.

Professor Block

Guide: **Professor Block** at world coordinates (4, 1, 4). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Welcome! Volume = Length $\times$ Width $\times$ Height. Explore the structures and solve problems!

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

How does your evidence support your result? Check it with a second method or describe a useful revision.

My mathematical reasoning and revisions

Quiz Master · Questions 1–2

Guide: **Quiz Master** at world coordinates (12, 5, 3). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 1 · Step 1

How many blocks LONG is this prism?

- A. 5 blocks B. 3 blocks C. 4 blocks

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 2 · Step 2

Good! How many blocks WIDE?

- A. 2 blocks B. 3 blocks C. 5 blocks

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Quiz Master · Questions 3–4

Guide: **Quiz Master** at world coordinates (12, 5, 3). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 3 · Step 3

And how many blocks **TALL**?

- A. 3 blocks B. 5 blocks C. 4 blocks

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 4 · Step 4

Now multiply: $5 \times 3 \times 4 = ?$

- A. 60 cubic units B. 12 cubic units C. 35 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Builder Bot · Questions 5–6

Guide: **Builder Bot** at world coordinates (4, 1, 11). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 5 · Step 1

What is the area of the pool floor? (3×3)

- A. 12 square units B. 9 square units C. 6 square units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 6 · Step 2

Now stack 2 layers of 9. How many total blocks?

- A. 11 blocks B. 27 blocks C. 18 blocks

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

L-Block Sage · Questions 7–8

Guide: **L-Block Sage** at world coordinates (18, 4, 12). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 7 · Step 1

Part A is $5 \times 3 \times 3$. What is its volume?

- A. 45 cubic units B. 15 cubic units C. 30 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 8 · Step 2

Part B is $3 \times 3 \times 3$. What is its volume?

- A. 9 cubic units B. 27 cubic units C. 18 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

L-Block Sage · Questions 9

Guide: **L-Block Sage** at world coordinates (18, 4, 12). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 9 · Step 3

Total volume = $45 + 27 = ?$

A. 45 cubic units B. 54 cubic units C. 72 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Connect the ideas

Choose one model or problem from this lesson. Explain how its representation connects to your calculations.

Model / diagram · show cubes, layers, dimensions, or a decomposition that fits the task

First calculation strategy · show the steps and units

Check with another strategy · explain why it agrees or what you revised

Reflection · what became easier and what do you still want to investigate?